MambaRefine-CD: Region-Boundary Temporal Refinement with MambaVision for Remote Sensing Change Detection

Abstract—**Remote-sensing change detection requires spatially coherent change masks and accurate boundary delineation. Existing CNN and transformer-based Siamese architectures model temporal differences effectively, but region-level change extent and boundary-level contour evidence are typically fused into a single representation, which can limit recovery of fine object boundaries. This paper presents MambaRefine-CD, a MambaVision-based binary change detection framework that explicitly decouples temporal evidence into complementary region-aware and boundary-aware responses. A shared MambaVision encoder extracts aligned multi-scale features from bi-temporal image pairs. The proposed Differential Region–Boundary Interaction (D-RBI) module applies separate bounded gates to a temporal feature stream constructed from both absolute and signed bi-temporal differences, producing distinct region and boundary pathways before multi-scale aggregation. An adaptive receptive-field decoder aggregates multi-scale context through parallel dilated convolution branches with per-image learned attention, and a bounded residual refinement head adjusts coarse predictions near boundaries with a controlled correction magnitude. On DSIFN-CD, averaged over two canonical runs, the method achieves F1 = 95.67 and IoU = 91.71. On WHU-CD, it achieves F1 = 95.15 and IoU = 90.76, using 65.40M parameters. A controlled ablation over seven variants (A0–A6) isolates the contribution of each component.**

I. Introduction

Remote-sensing change detection (CD) estimates a binary change mask from two co-registered images acquired over the same area at different times. The task underpins urban expansion monitoring, disaster assessment, land-cover analysis, and infrastructure management. In practice, binary CD is difficult because the model must compare two observations while being invariant to nuisance variation from illumination shifts, seasonal changes, shadows, sensor differences, and small registration errors. These factors are particularly visible near object boundaries, where a model can correctly identify the approximate changed region and still produce a poor mask if the predicted boundary is shifted, fragmented, or over-smoothed.

Deep CD methods have evolved along three main directions. Convolutional Siamese networks compare bi-temporal images through early fusion or feature differencing, as in FC-EF, FC-Siam-conc, and FC-Siam-diff [1]. Later CNN encoder–decoder models improve localization through skip connections, dense feature reuse, and multi-scale attention [2]–[4]. These models are effective because convolutions provide useful locality and translation bias, but their context modeling is bounded by the receptive field, which can limit reasoning about spatially distributed changes. Transformer-based methods address long-range interaction more directly: BIT models bi-temporal tokens with a Transformer encoder [5], and ChangeFormer uses a hierarchical Siamese Transformer for dense CD [6]. Wijenayake et al. [7] further demonstrate that precise spatio-temporal feature fusion yields gains over standard differencing strategies. Despite the progress, attention-based models carry high memory and compute cost at dense resolution, and stronger global representations do not automatically resolve boundary localization errors.

Recent visual state-space models (SSMs) provide a different route. Mamba introduces selective state-space sequence modeling with linear complexity [8], and VMamba adapts it to vision through two-dimensional selective scanning [9]. MambaVision develops a hybrid Mamba–Transformer backbone for visual recognition and dense prediction tasks [10]. Mamba-based CD models, including ChangeMamba [11], HAM-CD [12], and Mamba-CD [13], show that SSM backbones are competitive for binary and semantic change detection. Wijenayake et al. in Mamba-FCS [14] further demonstrate that combining frequency-domain fusion with change-guided attention and a SeK-inspired loss improves semantic CD. However, applying a Mamba-based encoder alone is not a complete solution. A CD model must still construct temporal evidence from both images and preserve fine boundaries in the output mask. A controlled benchmark of SSM backbones for remote-sensing segmentation by Wasalathilaka et al. [15] found that boundary delineation remains the dominant failure mode under domain shift, suggesting that encoder scaling alone is insufficient for boundary-sensitive tasks.

We propose **MambaRefine-CD**, a MambaVision-based architecture for binary remote-sensing CD. The model builds temporal features using raw paired features, absolute difference, and signed difference, giving the decoder access to both magnitude-based and direction-aware evidence. The core novelty is **D-RBI**, which applies two separate learned gating mechanisms to temporal features before decoding: a region gate that produces stable, coherent changed-area responses, and a Sobel-conditioned boundary gate that selects contour-sensitive evidence. These two pathways are combined through an adaptive receptive-field decoder and refined by a bounded residual head that applies a controlled correction to coarse logits near boundaries.

The contributions of this paper are:

- We present **MambaRefine-CD**, a MambaVision-based binary CD architecture integrating shared bi-temporal encoding, explicit four-stream temporal feature construction, and boundary-aware refinement.

- We propose **D-RBI**, a lightweight module that decouples temporal evidence into complementary region-aware and boundary-aware gated pathways before multi-scale decoding.
- We integrate adaptive receptive-field decoding, CRAM-lite spatial modulation, and bounded residual boundary refinement into a unified pipeline, demonstrating that each component contributes measurably.
- We report verified results on DSIFN-CD and WHU-CD, averaged over independent runs, and present a controlled ablation study over seven variants (A0–A6) that isolates each component's effect on F1, IoU, and OA.

II. Related Work

A. Convolutional and Transformer-Based Change Detection

Early deep learning CD models used convolutional Siamese encoders with feature differencing or concatenation as the temporal comparison mechanism [1]. Dense decoders with skip connections and multi-scale aggregation improved spatial localization [2]–[4]. These CNN-based approaches are efficient and translation-equivariant, but their effective receptive field limits the ability to relate spatially distant changed regions.

Transformer-based methods address global context more directly. BIT encodes bi-temporal features as semantic tokens and applies cross-attention for temporal reasoning [5]. ChangeFormer uses a hierarchical Siamese Transformer with an MLP decoder for dense prediction [6]. Ratnayake et al. show that incorporating attention and improved loss design into segmentation architectures benefits change detection tasks [16]. While transformers improve global modeling, their quadratic attention complexity makes high-resolution binary CD computationally expensive, and improved global representations do not directly translate to better boundary delineation.

B. State Space Models for Remote Sensing

Mamba introduces selective SSM sequence modeling with linear complexity [8]. VMamba extends this to vision via two-dimensional selective scanning [9]. MambaVision develops a hybrid Mamba–Transformer backbone with competitive accuracy and efficiency on dense prediction benchmarks [10].

In CD, ChangeMamba integrates VMamba into a bi-temporal architecture and demonstrates competitive performance [11]. HAM-CD introduces a hybrid attention mechanism to compensate for Mamba's limitations in fine-grained local modeling [12]. Wijenayake et al. combine spectral-domain fusion with change-guided attention and a SeK-inspired loss in Mamba-FCS for semantic CD [14], while Wijenayake et al. further explore precise spatio-temporal fusion strategies for binary CD [7].

Peng et al. proposed Mamba-CD [17], a VMamba-based binary change detection network with change region-aware attention and a gated recursive context refinement mechanism. Their work is closely related to MambaRefine-CD because it also studies Mamba-style long-range modeling and boundary-sensitive binary change detection. However, Mamba-CD enhances shallow features using a change-region attention map and recursively fuses multiscale context, whereas MambaRefine-CD explicitly constructs temporal evidence using paired features, absolute difference, and signed difference, and separates region and boundary streams through D-RBI before bounded boundary residual refinement.

These works demonstrate the viability of SSM backbones for remote-sensing CD, but they primarily address region-level detection and do not explicitly separate boundary-sensitive cues from region-level evidence.

C. Boundary-Aware Modeling and Temporal Feature Construction

Boundary precision is a well-recognized challenge in segmentation. Boundary loss directly penalizes errors at object contours [18]. Gated-SCNN processes a dedicated shape stream in parallel with the main segmentation path [19]. BASNet uses a two-stage coarse-then-fine architecture for salient object boundary detection [20]. In remote sensing, boundary-aware multi-task learning has been studied for aerial dense prediction [21], and BSCNet uses boundary-aware semantic context to sharpen CD edge predictions [22].

A key limitation of most CD methods, including recent Mamba adaptations, is that they compare bi-temporal features through absolute differencing $|F_2 - F_1|$ alone. Absolute differencing is symmetric and discards the temporal direction of the change (e.g., whether a structure was built or demolished). Wasalathilaka et al. confirm that boundary delineation remains the dominant failure mode of SSM backbones under domain shift, even at larger scales [15]. MambaRefine-CD addresses both issues: it preserves signed temporal direction through $F_2 - F_1$ and explicitly gates the temporal feature into region and boundary pathways before decoding.

III. Methodology

A. Overview

MambaRefine-CD predicts a binary change map from a bi-temporal image pair (I_1, I_2) through five components: a shared MambaVision encoder, a four-stream temporal feature constructor, D-RBI, an adaptive receptive-field FPN decoder with CRAM-lite modulation, and a bounded boundary residual refinement head. Fig. **??** shows the overall pipeline.

B. Shared MambaVision Encoding

A shared encoder $E(\cdot)$ processes both input images in the same feature space:

$$\{F_1^s\}_{s=1}^S = E(I_1), \qquad \{F_2^s\}_{s=1}^S = E(I_2). \tag{1}$$

Weight sharing ensures that the temporal difference computed downstream reflects genuine change rather than encoder domain mismatch. We use MambaVision-S as the default backbone (channels $[96, 192, 384, 768]$, $S = 4$ stages), initialized from pretrained weights and kept trainable. The backbone fuses Mamba layers for long-range modeling and Transformer layers for local attention [10].

C. Four-Stream Temporal Feature Construction

At each encoder stage s, we apply independent GroupNorm to F_1^s and F_2^s, then concatenate four streams:

$$T^s = \big[F_{1,n}^s,\ F_{2,n}^s,\ |F_{2,n}^s - F_{1,n}^s|,\ F_{2,n}^s - F_{1,n}^s\big], \qquad (2)$$

where $F_{i,n}^s$ denotes the normalized feature. Absolute difference provides a symmetric change-magnitude cue; signed difference preserves the temporal direction of change, which conveys whether a structure appeared or disappeared. This four-stream concatenation T^s is projected to a fixed channel width through a lightweight 1×1 convolution followed by a depthwise 3×3 and pointwise 1×1 block, yielding the temporal descriptor D^s.

D. Differential Region–Boundary Interaction (D-RBI)

D-RBI applies two separate gating mechanisms to D^s to expose region-level and boundary-level evidence before decoding. Fig. **??** illustrates the module.

Region gate. A spatial attention path processes D^s through a depthwise 3×3 convolution and a pointwise 1×1 convolution to produce unnormalized gate logits, which are clamped to $[-8, 8]$ and passed through sigmoid, then scaled to bound the response:

$$G_r^s = 0.6\cdot\sigma\big(\mathrm{clamp}(g_r(D^s), -8, 8)\big)+0.2, \qquad G_r^s \in [0.2, 0.8]. \qquad (3)$$

The lower bound prevents complete feature suppression; the upper bound prevents unrestricted amplification.

Boundary gate. A Sobel-conditioned gate modulates its response using the magnitude of image-level edges computed on D^s. Sobel magnitude is clamped to $[0, 10]$ and normalized, providing a spatial prior on likely boundary locations:

$$G_b^s = 0.4\cdot\sigma\big(S^s\cdot g_b(D^s)\big), \qquad G_b^s \in [0.0, 0.4], \qquad (4)$$

where S^s denotes the normalized Sobel magnitude map. The tighter bound keeps the boundary gate from dominating when edge evidence is unreliable.

Outputs. The two gated responses and the original descriptor are returned:

$$R^s = G_r^s \odot D^s, \qquad B^s = G_b^s \odot D^s, \qquad D^s \text{ (unchanged)}. \qquad (5)$$

These three streams give the decoder access to region evidence, boundary evidence, and unmodulated temporal features.

E. CRAM-Lite Modulation and Adaptive Receptive-Field Decoding

At selected encoder stages ($s \in \{0, 1, 2\}$), a lightweight CRAM-lite block reweights the region features spatially. Following the motivation of change-region-aware attention in Mamba-CD [13], each CRAM-lite block estimates a per-location attention map $A \in [0, 1]$ via depthwise and pointwise convolutions and applies a residual modulation:

$$F_{\mathrm{out}} = F_{\mathrm{region}}\cdot(1+\alpha\cdot A), \qquad \alpha \in \mathbb{R}^+ \text{ (learnable, init 0.5)}. \qquad (6)$$

The modulated multi-scale features are decoded through a top-down FPN with adaptive receptive-field blocks. Each block runs four parallel dilated depthwise convolutions with rates $\{1, 2, 4, 8\}$, weights each branch via per-image softmax attention (estimated through global average pooling and a shared linear layer), and sums the weighted outputs. This gives the decoder access to both compact object-level changes and larger spatial structures without altering the input resolution.

F. Coarse Head and Bounded Boundary Residual Refinement

The decoder first produces a coarse logit map P_c through a 3×3 convolution followed by a 1×1 output projection, bilinearly upsampled to the input resolution. A boundary refinement head then computes a bounded correction:

$$P_f = P_c + 0.1\cdot\tanh(\Delta), \qquad (7)$$

where Δ is estimated from the concatenation of the coarsest boundary feature B^0, the coarse logit P_c, and Sobel edges of $\sigma(P_c)$. The $\tanh$ bounds the correction magnitude; the scalar 0.1 limits refinement to uncertain edge regions without freely overwriting the coarse global prediction. The final linear layer of the boundary head is zero-initialized to ensure the model begins training from the coarse prediction alone.

G. Training Objective

The full training loss is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}} + 0.4\,\mathcal{L}_{\mathrm{coarse}} + 0.1\,\mathcal{L}_{\mathrm{boundary}}, \qquad (8)$$

where $\mathcal{L}_{\mathrm{BCE}}$ and $\mathcal{L}_{\mathrm{Dice}}$ supervise the final output P_f; $\mathcal{L}_{\mathrm{coarse}}$ applies the same binary supervision to P_c; and $\mathcal{L}_{\mathrm{boundary}}$ is a Sobel-edge ℓ_1 loss between predicted and ground-truth boundaries. Ablation variants A0–A5 are trained with only $\mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}}$ on the final output (coarse and boundary loss weights set to zero).

IV. Experiments

A. Datasets and Split Protocol

We evaluate MambaRefine-CD on two widely used binary CD benchmarks.

DSIFN-CD [23] contains 394 pairs of Gaofen-2 satellite images at 1 m GSD covering six Chinese cities with complex land-cover transitions. We use the official image-level split files (395 train, 48 validation, 48 test image pairs), strictly enforcing non-overlap between splits. Patches are extracted from training images only, and test images are evaluated in patch mode with 25% overlap.

WHU-CD [24] is an aerial building CD dataset collected over Christchurch, New Zealand. We use the official three-way split (train / validation / test) and evaluate test images in patch mode with the same patch configuration.

B. Implementation Details

Models are trained with AdamW (lr$= 5 \times 10^{-5}$, weight decay$= 0.01$), cosine scheduling with 2,500 warmup iterations, batch size 8, and image crops of 256×256. Gradient norm is clipped at 0.5. Exponential moving average (EMA, decay$= 0.999$) is maintained throughout training; test inference uses EMA weights. Augmentation consists of random horizontal and vertical flips. Training uses 16-bit automatic mixed precision on a single NVIDIA Quadro GV100 (32 GB). Ablation variants A0–A5 are trained for 30,000 iterations; the full model (A6) is trained for 50,000 iterations. Validation is performed every 5,000 iterations and the checkpoint with the best validation F1 is selected. For threshold selection, we sweep $\{0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60\}$ on the validation set and fix the chosen threshold for test evaluation; test metrics are never used for selection.

C. Evaluation Metrics

We report Precision (Pre), Recall (Rec), F1, Intersection-over-Union (IoU), and Overall Accuracy (OA), all computed for the changed class using a global confusion matrix over the full evaluation set. Results are averaged over independent training runs where multiple runs are available.

OA is strongly affected by the dominant unchanged class. The lower DSIFN-CD OA relative to some literature values is not directly comparable because our evaluation uses a different clean split. We therefore emphasise changed-class F1 and IoU as primary metrics, while reporting OA for completeness.

In addition to standard pixel-level metrics, we evaluate three boundary-specific metrics: (i) Boundary F1 (BF1), the tolerance-aware F1 between predicted and GT boundary pixels at a 3-pixel tolerance; (ii) Boundary IoU (BIoU), the IoU between 3-pixel-dilated boundary bands of the prediction and GT; and (iii) Trimap F1 at 3 px, the changed-class F1 restricted to the 3-pixel band around the GT boundary. These metrics provide direct evidence of boundary localisation quality, which is the primary claim of MambaRefine-CD.

D. Main Results

Table I reports the full model performance on both datasets. MambaRefine-CD achieves F1 = 95.67 and IoU = 91.71 on DSIFN-CD (averaged over two canonical runs) and F1 = 95.15 and IoU = 90.76 on WHU-CD, using 65.40M parameters. The best single DSIFN-CD run achieves F1 = 96.40 and IoU = 93.04, indicating that run-to-run variance is present and reporting averaged metrics is important for reliable comparison.

E. Boundary-Specific Evaluation

Table II reports boundary metrics across four ablation variants on DSIFN-CD and the WHU-CD full model. The full model (A6) consistently achieves the highest BF1 and Trimap F1-3px, confirming that the joint boundary head and Sobel-edge loss produce measurably better boundary localisation than the baseline. The boundary residual head alone (A5), without boundary-aware loss, does not improve BF1 over A4,

TABLE I
BINARY CHANGE DETECTION RESULTS. DSIFN-CD VALUES ARE AVERAGED OVER TWO CANONICAL RUNS. WHU-CD IS A SINGLE VERIFIED RUN AT THRESHOLD 0.55. THE MODEL USES MAMBAVISION-S WITH 65.40M PARAMETERS.

Dataset	Pre	Rec	F1	IoU	OA
DSIFN-CD	95.47	95.87	95.67	91.71	96.98
WHU-CD	95.58	94.74	95.15	90.76	99.54

TABLE II
BOUNDARY-SPECIFIC METRICS (3-PIXEL TOLERANCE). BF1 = BOUNDARY F1; BIOU = BOUNDARY BAND IOU; TRIMAP F1 = CHANGED-CLASS F1 IN THE 3-PX GT BOUNDARY BAND.

Variant	F1	IoU	BF1	BIoU	Trimap F1-3px
A1 MambaVision-FPN	93.21	87.28	61.36	38.41	68.02
A4 + ARF-FPN	93.71	88.16	65.94	42.06	70.27
A5 + Boundary Residual	93.59	87.94	65.14	41.37	69.98
A6 Full Model	94.58	89.72	71.94	47.39	72.93
WHU Full Model	95.15	90.76	89.49	71.37	87.01

which is consistent with the standard F1 regression observed in the ablation study.

F. Qualitative Boundary Analysis

Fig. 1 shows representative test samples from DSIFN-CD. Each row displays the two input images, ground-truth mask, full-model prediction, error map (white = TP, black = TN, red = FP, green = FN), and a boundary overlay showing GT boundaries in yellow and predicted boundaries in cyan. The full model (A6) produces sharper boundary delineation with fewer false positives at edge regions compared to the baseline, consistent with the quantitative boundary metrics.

G. Comparison with Recent Change Detection Methods

We compare MambaRefine-CD with recent CNN-, Transformer-, hybrid-, and Mamba-based binary change detection methods reported by Peng et al. [17]. The WHU-CD comparison follows the standard WHU test setting used in recent literature. For DSIFN-CD, we report the literature values from Peng et al. together with our verified clean-split result; because the DSIFN split differs (Peng et al. use a 14400/1360/192 patch-pair partition, while we use image-level non-overlapping splits), the DSIFN table should be interpreted as contextual rather than a strict same-protocol ranking.

Colored cells denote the best, second-best, and third-best values within each metric column.

On WHU-CD (Table III), MambaRefine-CD reaches 95.15% F1 and 90.76% IoU, closely matching the recent Mamba-CD result [17] while obtaining higher recall (94.74 vs. 95.61 for pre, 94.74 for rec). For DSIFN-CD, because the evaluation split differs across studies—Peng et al. [17] use a 14400/1360/192 patch-pair partition while we use image-level non-overlapping splits—a direct table comparison would be misleading. Instead we note that our averaged clean-split

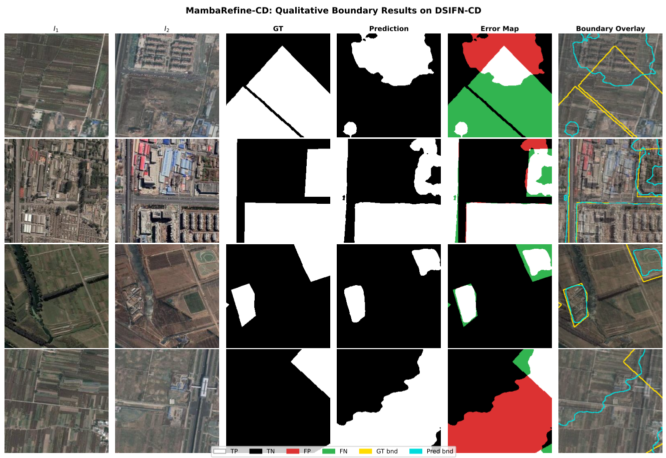

Fig. 1. Qualitative boundary results on DSIFN-CD (A6 full model). Each row: I_1, I_2, GT, prediction, error map (white = TP, black = TN, red = FP, green = FN), boundary overlay (yellow = GT bnd, cyan = pred bnd).

TABLE III
COMPARISON WITH RECENT METHODS ON WHU-CD. LITERATURE VALUES ARE REPORTED BY PENG ET AL. [17].

Method	Year	Pre	Rec	F1	IoU	OA
FC-EF [1]	2018	71.63	67.25	69.37	53.11	97.61
FC-Siam-Di [1]	2018	47.33	77.66	58.81	41.66	95.63
FC-Siam-Conc [1]	2018	60.88	73.58	66.63	49.95	97.04
DTCDSCN	2020	63.92	82.30	71.95	56.19	97.42
STANet	2020	79.37	85.50	82.32	69.95	98.52
IFNet	2020	**96.91**	73.19	83.40	71.52	98.83
SNUNet [3]	2021	85.60	81.49	83.50	71.67	98.71
ChangeFormer [6]	2022	91.83	88.02	89.88	81.63	99.12
BiFA	2024	95.15	**93.60**	**94.37**	**89.34**	**99.56**
RSM-CD	2024	93.37	90.42	91.87	84.96	–
SChanger	2025	94.62	91.83	93.20	87.27	–
CDMamba	2025	**95.58**	92.01	93.76	88.26	99.51
Mamba-CD [17]	2026	**96.52**	**93.91**	**95.20**	**90.83**	**99.62**
MambaRefine-CD (ours)	2026	**95.58**	**94.74**	**95.15**	**90.76**	**99.54**

result (F1 = 95.67, IoU = 91.71) is comparable to Mamba-CD (F1 = 95.61, IoU = 91.69) in absolute magnitude; this is contextual rather than a strict same-protocol ranking, and we make no SOTA claim on DSIFN-CD.

H. Ablation Study

Table IV presents a controlled ablation over seven variants on DSIFN-CD. Starting from a lightweight CNN baseline (A0, 7.84M, F1 = 76.63), replacing the encoder with MambaVision-S (A1) yields a large gain (+17.24 F1), confirming the encoder's dominant role. The A2 row is a single verified run using MambaVision-S; additional repeated runs are left for future work. Introducing the signed temporal difference (A3) validates the directional cue (+0.95 F1). Adding CRAM-lite modulation (A4) provides a further marginal improvement (+0.08 F1). Adding the boundary residual head alone without boundary-aware loss terms (A5) causes a regression (−0.77 F1). This is expected: boundary residual alone gives a small drop, but the full model improves when the boundary residual is trained jointly with CRAMLite, coarse auxiliary supervision, and boundary loss. The full model (A6) with all loss terms recovers and exceeds prior variants, achieving

TABLE IV
ABLATION STUDY ON DSIFN-CD. A0–A5 USE 30K ITERATIONS; A6 USES 50K. VALUES AVERAGED OVER AVAILABLE RUNS. PARAMS IN MILLIONS. A2 IS A SINGLE VERIFIED RUN; ADDITIONAL RUNS ARE LEFT FOR FUTURE WORK.

ID	Variant	F1	IoU	OA	Params
A0	CNN baseline	76.63	62.12	83.80	7.84
A1	+ MambaVision-S	93.87	88.45	95.72	53.54
A2	+ D-RBI (abs only)	93.33	87.50	95.35	54.98
A3	+ Signed diff	94.28	89.19	95.99	55.34
A4	+ CRAM-lite	94.36	89.32	96.05	65.12
A5	+ Bnd. residual head	93.59	87.94	95.53	65.19
A6	Full model (+ bnd. loss)	**95.67**	**91.71**	**96.98**	65.40

F1 = 95.67. This confirms that the boundary head and the Sobel-edge loss term must be enabled together.

I. Backbone Scaling

Moving from the tiny backbone (46.73M, F1 = 94.58) to small (65.40M, F1 = 95.67) yields a clear +1.09 F1 gain. Moving to base (113.37M) gives only +0.10 F1 at 74% more parameters, suggesting diminishing returns beyond small.

J. Limitations

The Sobel-based boundary gate provides a useful spatial prior but is not learned end-to-end; replacing it with a learned edge detector could improve performance under complex textures. DSIFN-CD and WHU-CD protocols differ across studies, making strict cross-paper comparisons on DSIFN-CD unreliable; future work should standardise the evaluation split. Additionally, the small-to-base backbone scaling shows diminishing returns, suggesting that architecture improvements beyond encoder scaling are needed for further gains.

V. CONCLUSION

This paper presented MambaRefine-CD, a MambaVision-based architecture for binary remote-sensing change detection. The method decouples temporal evidence into region-aware and boundary-aware pathways through D-RBI, combines signed temporal difference with adaptive receptive-field decoding and CRAM-lite spatial modulation, and applies a bounded residual correction near boundaries. A controlled ablation over seven variants demonstrates that the MambaVision encoder provides the dominant accuracy gain, while signed temporal difference, CRAM-lite modulation, and the combined boundary head plus Sobel-edge loss each contribute additional improvement. On DSIFN-CD (averaged over two runs) the method achieves F1 = 95.67 and IoU = 91.71. On WHU-CD, it achieves F1 = 95.15 and IoU = 90.76. The results show that MambaRefine-CD is competitive with recent Mamba-based change detection methods, including Mamba-CD [17], while providing a different region-boundary temporal refinement design. Future work will extend evaluation to additional binary CD benchmarks, investigate learned boundary gate designs beyond the Sobel prior, and explore applying the region–boundary decomposition to semantic CD settings.

References

[1] R. C. Daudt, B. Le Saux, and A. Boulch, “Fully convolutional siamese networks for change detection,” in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2018, pp. 4063–4067.

[2] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.

[3] S. Fang, K. Li, J. Shao, and Z. Li, “Snunet-cd: A densely connected siamese network for change detection of vhr images,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[4] J. Chen, Z. Yuan, J. Peng, L. Chen, H. Huang, J. Zhu, Y. Liu, and H. Li, “Dasnet: Dual attentive fully convolutional siamese networks for change detection in high-resolution satellite images,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 1194–1206, 2021.

[5] H. Chen, Z. Qi, and Z. Shi, “Remote sensing image change detection with transformers,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2022.

[6] W. G. C. Bandara and V. M. Patel, “A transformer-based siamese network for change detection,” in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022, pp. 207–210.

[7] W. Wijenayake, R. Ratnayake, D. Sumanasekara, N. Wasalathilaka, M. Piratheepan, G. Godaliyadda, M. Ekanayake, and H. Herath, “Precision spatio-temporal feature fusion for robust remote sensing change detection,” in *2025 IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, vol. 19, 2026, pp. 557–562.

[8] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.

[9] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, J. Jiao, and Y. Liu, “Vmamba: Visual state space model,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[10] A. Hatamizadeh and J. Kautz, “Mambavision: A hybrid mamba-transformer vision backbone,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[11] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, “Changemamba: Remote sensing change detection with spatiotemporal state space model,” *IEEE Transactions on Geoscience and Remote Sensing*, 2024.

[12] G. Li, P. Han, W. Wang, T. Mu, Z. Xiao, and X. Li, “Ham-cd: Hybrid attention mamba for remote sensing change detection,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 64, pp. 1–18, 2026.

[13] G. Peng, Y. Li, J. Chen, and Y. Zhang, “Mamba-cd: Mamba-based change detection network for remote sensing images with change region-aware attention and recursive context refinement mechanism,” *IEEE Transactions on Geoscience and Remote Sensing*, 2026.

[14] B. Wijenayake, A. Ratnayake, P. Sumanasekara, R. Godaliyadda, P. Ekanayake, V. Herath, and N. Wasalathilaka, “Mamba-fcs: Joint spatio-frequency feature fusion, change-guided attention, and sek inspired loss for enhanced semantic change detection in remote sensing,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 19, pp. 7680–7698, 2026.

[15] N. Wasalathilaka, D. Perera, O. Samarakoon, B. Wijenayake, R. Godaliyadda, V. Herath, and P. Ekanayake, “A controlled benchmark of visual state-space backbones with domain-shift and boundary analysis for remote-sensing segmentation,” *arXiv preprint arXiv:2604.18721*, 2026.

[16] R. Ratnayake, W. Wijenayake, D. Sumanasekara, G. Godaliyadda, H. Herath, and M. Ekanayake, “Enhanced scannet with cbam and dice loss for semantic change detection,” in *2025 Moratuwa Engineering Research Conference (MERCon)*, 2025, pp. 84–89.

[17] A. Peng, L. Zhuo, L. Wang, H. Zhang, and H. Xu, “Mamba-cd: Mamba-based change detection network for remote sensing images with change region-aware attention and recursive context refinement mechanism,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 19, pp. 4288–4300, 2026.

[18] H. Kervadec, J. Bouchtiba, C. Desrosiers, E. Granger, J. Dolz, and I. B. Ayed, “Boundary loss for highly unbalanced segmentation,” in *International Conference on Medical Imaging with Deep Learning (MIDL)*, 2019, pp. 285–296.

[19] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, “Gated-scnn: Gated shape cnns for semantic segmentation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 5229–5238.

[20] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O. R. Zaiane, and M. Jagersand, “Basnet: Boundary-aware salient object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 7479–7489.

[21] Y. Wang, W. Ding, R. Zhang, and H. Li, “Boundary-aware multi-task learning for remote sensing imagery,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 951–963, 2021.

[22] W. Zhou, G. Guo, H. Song, X. Zhang, and K. Zhang, “Learning boundary-aware semantic context network for remote sensing change detection,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 19, pp. 4177–4188, 2026.

[23] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, “Deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images,” *Information Fusion*, vol. 83–84, pp. 43–58, 2022.

[24] S. Ji, S. Wei, and M. Lu, “Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 1, pp. 574–586, 2019.